

1. Define vital capacity. What is its significance?

☞ Vital capacity is the volume of air that can breathe in after a forced expiration or Volume of air that can breathe out after a forced inspiration.

Significance: Greater the vital capacity more is the energy available to the body.

2. State the volume of air remaining in the lungs after a normal breathing.

☞ Functional residual capacity (FRC). $FRC = RV + ERV = 2200 - 2300$ ml.

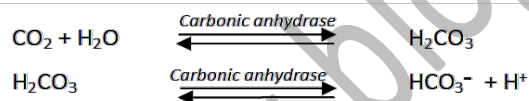
3. Diffusion of gases occurs in the alveolar region only and not in the other parts of respiratory system. Why?

☞ For effective gas exchange, respiratory surface must be thin, highly vascular and with large surface area. Only alveolar region has these features.

4. What are the major transport mechanisms for CO₂? Explain.

☞ **CO₂ transport** occurs in 3 ways:

- **As carbonic acid:** In tissues, about **7%** of CO₂ is dissolved in **plasma water** to form **carbonic acid** and carried to lungs.
- **As carbamino-haemoglobin:** In tissues, **20-25%** of CO₂ binds to Hb to form **carbamino-haemoglobin**. In alveoli, CO₂ dissociates from carbamino-haemoglobin.
- **As bicarbonates:** About **70%** of CO₂ is transported by this method. RBCs and plasma contain an enzyme, *carbonic anhydrase*. This enzyme facilitates the following reactions.

**5. What will be the pO₂ and pCO₂ in the atmospheric air compared to those in the alveolar air?**

- (i) pO₂ lesser, pCO₂ higher
- (ii) pO₂ higher, pCO₂ lesser
- (iii) pO₂ higher, pCO₂ higher
- (iv) pO₂ lesser, pCO₂ lesser.

☞ pO₂ higher, pCO₂ lesser.

6. Explain the process of inspiration under normal conditions.

☞ During inspiration, the **diaphragm** contracts (flattens) causing an increase in vertical volume (*antero-posterior axis*).

Contraction of **external intercostal muscles** (muscles found between ribs) lifts up the ribs and sternum causing an increase in thoracic volume in the *dorso-ventral axis*.

These changes reduce pressure inside the thorax causing the expansion of lungs. Thus pulmonary volume increases resulting in decrease of *intra-pulmonary pressure* to less than the atmospheric pressure. So air moves into lungs.

7. How is respiration regulated?

☞ Respiratory centres present in the brain include

- **Respiratory rhythm centre (Inspiratory & Expiratory centres):** In **medulla oblongata**.
- **Pneumotaxic centre:** In **Pons**. It moderates the functions of the respiratory rhythm centre. Neural signal from this centre reduces the duration of inspiration and thereby alter the respiratory rate.
- **Chemosensitive area:** Seen adjacent to the rhythm centre. Increase in the concentration of CO₂ and H⁺ activates this centre, which in turn signals rhythm centre. **Receptors** in **aortic arch** and **carotid artery** also recognize changes in CO₂ and H⁺ concentration and send signals to the rhythm centre.

8. What is the effect of pCO₂ on oxygen transport?

- ☞ Increase in $p\text{CO}_2$ in blood causes rightward shift of the oxygen-haemoglobin dissociation curve. It decreases the affinity of Hb for oxygen (Bohr's effect). It plays a crucial role in enhancing oxygenation of blood in the lungs and release of oxygen in the tissues.

9. What happens to the respiratory process in a man going up a hill?

- ☞ His breathing rate will increase.

10. What is the site of gaseous exchange in an insect?

- ☞ Tracheae.

11. Define oxygen dissociation curve. Can you suggest any reason for its sigmoidal pattern?

- ☞ Oxygen dissociation curve is a sigmoid curve obtained when percentage saturation of Hb with O_2 is plotted against the $p\text{O}_2$.

The sigmoidal pattern is due to following properties:

- Minimal loss of O_2 from haemoglobin occurs above $p\text{O}_2$ of 70-80 mm Hg despite significant changes in tension of oxygen beyond this. This is depicted by flat portion of the curve.
- Significant change of the dissociation curve below $p\text{O}_2$ of 40 mm Hg ensures that with any further decline in $p\text{O}_2$ there will be a greater release of oxygen from haemoglobin.

12. Have you heard about hypoxia? Try to gather information about it, and discuss with your friends.

- ☞ Hypoxia is a condition of oxygen shortage in the tissues. It is 2 types:
- Artificial hypoxia:** It is due to shortage of oxygen in the air as at high altitude. It causes mountain sickness. Its symptoms are breathlessness, headache, dizziness and bluish tinge on skin.
 - Anaemic hypoxia:** It is due to the reduced oxygen carrying capacity of the blood due to anaemia or carbon monoxide poisoning. Here, less haemoglobin is available for carrying O_2 .

13. Distinguish between

- IRV and ERV
- Inspiratory capacity and Expiratory capacity.
- Vital capacity and Total lung capacity.

- ☞ (a) **Inspiratory reserve volume (IRV)** is an additional volume of air that can inspire by forceful inspiration.
Expiratory reserve volume (ERV) is an additional volume of air that can expire by a forceful expiration.
- (b) **Inspiratory capacity (IC)** is the total volume of air inspired after a normal expiration ($\text{TV} + \text{IRV}$).
Expiratory capacity (EC) is the total volume of air expired after a normal inspiration ($\text{TV} + \text{ERV}$).
- (c) **Vital capacity (VC)** is the volume of air that can breathe in after a forced expiration or Volume of air that can breathe out after a forced inspiration ($\text{ERV} + \text{TV} + \text{IRV}$).
Total lung capacity (TLC) is the total volume of air in the lungs after a maximum inspiration. ($\text{RV} + \text{ERV} + \text{TV} + \text{IRV}$ or $\text{VC} + \text{RV}$).

14. What is Tidal volume? Find out the Tidal volume (approximate value) for a healthy human in an hour.

- ☞ **Tidal volume (TV)** is the volume of air inspired or expired during a normal respiration. It is about **500 ml**.
 $\text{TV for a healthy human per minute} = 500 \times \text{breathing rate (12-16)} = 6000 \text{ to } 8000 \text{ ml.}$
 $\text{Hence TV in an hour} = 6000 \text{ to } 8000 \times 60 = 360000 - 480000 \text{ ml.}$